

1.Problem

Title

Water Jug Problem (4-Gallon and 3-Gallon Jugs)

Problem Statement

You are given two water jugs: one with a capacity of 4 gallons and another with a capacity of 3 gallons. Neither jug has any measurement markings. A water pump is available to fill either jug completely. Water can also be poured from one jug to the other until either the source jug is empty or the destination jug is full. Water may also be emptied onto the ground.

The objective is to obtain exactly 2 gallons of water in the 4-gallon jug.

Assumptions

- The 4-gallon jug can hold a maximum of 4 gallons.
- The 3-gallon jug can hold a maximum of 3 gallons.
- Both jugs are initially empty.
- Water can be filled completely from the pump.
- Water can be poured from one jug to another.
- Water can be discarded onto the ground.
- No other measuring devices are available.

2. Mars Rover – Analyze and Explore the Samples :

Problem

Title

Mars Rover – Intelligent Agent

Problem Statement

A Mars Rover is an intelligent robotic agent developed to explore the surface of Mars without continuous human intervention. The rover collects soil, rock, and atmospheric samples, captures images, analyzes the surrounding environment, and sends scientific data back to Earth. Since communication between Earth and Mars has a significant time delay, the rover must make many decisions autonomously.

The objective is to analyze the Mars Rover as an Intelligent Agent by identifying:

- The percepts received by the agent.
- The characteristics of its operating environment.
- The actions performed by the rover.

- The performance measures used to evaluate it.
- The most suitable intelligent agent architecture and the reason for choosing it.

3.8-Queens Problem using Search Space

Problem

Title

8-Queens Problem using Search Space

Problem Statement

The 8-Queens Problem is a classical Artificial Intelligence search problem. The objective is to place eight queens on an 8×8 chessboard such that no two queens attack each other. A queen can attack another queen if they are in the same row, same column, or same diagonal.

The goal is to find a valid arrangement of all eight queens using a suitable search strategy. The search space consists of all possible board configurations, and the algorithm explores these configurations until a valid solution is found.

Objective

- Place eight queens on the chessboard.
- Ensure no two queens share the same row, column, or diagonal.
- Find a valid solution using a search algorithm.

4. OLA Cab Booking – Intelligent Agent

Problem

Title

OLA Cab Booking – Intelligent Agent

Problem Statement

OLA Cab Booking is an intelligent transportation system that helps customers book a suitable cab to travel from one location to another. The customer enters the pickup and destination locations and selects a preferred cab type such as Mini, Micro, Sedan, Shared, Prime, or any other available option. The intelligent agent processes the user's request, checks the availability of drivers, estimates the fare and arrival time, and assigns the most suitable cab. The objective is to provide a safe, fast, and cost-effective ride while minimizing waiting time.

Objective

- Accept the customer's booking request.
- Identify the pickup and destination locations.
- Check the availability of cab types.

- Allocate the most suitable cab.
- Confirm the booking and display ride details.

5.Uniform Cost Search (UCS)

Problem

Title

Uniform Cost Search (UCS) for Logistics Delivery Network

Problem Statement

A logistics company operates a delivery network in which packages are transported between warehouses. Each route between two warehouses has an associated travel cost based on fuel consumption and travel time. The company aims to determine the least-cost path from the starting warehouse (S) to the destination warehouse (G) using the Uniform Cost Search (UCS) algorithm.

Uniform Cost Search is an informed search algorithm that always expands the node with the lowest cumulative path cost, ensuring that the optimal (least-cost) path is found.

Objective

- Represent the delivery network as a weighted graph.
- Explore the search space using Uniform Cost Search.
- Find the least-cost path from **S** to **G**.
- Display the optimal path and total travel cost.